Assignment #4

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Problem D.4, D.6

Find:
- Equilibria. (eqba.)
- Jacobian linearization. (J.L.)
- Feedback linearization (F.L.)

- For $\vec{x} = \begin{bmatrix} z \\ \dot{z} \end{bmatrix}$ and $\vec{u} = [F]$ and $\vec{y} = [z]$, find state space form (SSF):

$$\dot{\vec{x}} = A\vec{x} + B\vec{u}$$
$$\vec{y} = C\vec{x} + D\vec{u}$$

Solution

From Assignment #3, we have

$$\dot{\vec{x}} = \vec{f}(x, u) = \begin{bmatrix} d\,z/dt \\ \frac{d}{dt}(\dot{z}) \end{bmatrix} = \begin{bmatrix} \dot{z} \\ \frac{1}{m}(F - b\dot{z} - kz) \end{bmatrix}$$

Then for $f(x_e, u_e) = 0 \ldots$

$$\dot{z}_e = 0$$

$$F_e - b\cancel{\dot{z}_e}^{\,0} - k z_e = 0 \Rightarrow F_e = k z_e$$

Therefore the equilibria are coupled such that...

$$\left.\begin{aligned} \dot{z}_e &= 0 \\ F_e &= k z_e \end{aligned}\right\}$$ equilibrium

All other state variables (z) are unconstrained.

Note the EoMs are already linear, so linearization is unnecessary. JL FL

By inspection...

$$\left.\begin{aligned} \dot{\vec{x}} &= \begin{bmatrix} 0 & 1 \\ -k/m & -b/m \end{bmatrix} \begin{bmatrix} z \\ \dot{z} \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} [F] \\ \vec{y} &= \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} z \\ \dot{z} \end{bmatrix} + [0][F] \end{aligned}\right\}$$ SSF

Problem E.4, E.6

Find: Someus D, with $\vec{\tilde{x}} = \begin{bmatrix} \tilde{z} \\ \tilde{\theta} \\ \dot{\tilde{z}} \\ \dot{\tilde{\theta}} \end{bmatrix}$ and $\vec{\tilde{u}} = [\tilde{F}]$

and $\vec{\tilde{y}} = \begin{bmatrix} \tilde{z} \\ \tilde{\theta} \end{bmatrix}$

From Assignment #3...

$$\dot{\vec{x}} = \vec{f}(\vec{x}, \vec{u}) = \begin{bmatrix} \frac{d}{dt}(z) \\ \frac{d}{dt}(\theta) \\ \frac{d}{dt}(\dot{z}) \\ \frac{d}{dt}(\dot{\theta}) \end{bmatrix} = \begin{bmatrix} \dot{z} \\ \dot{\theta} \\ -g\sin\theta + z\dot{\theta}^2 \\ \frac{3((2Fl - m_2 g l - 2m_1 g z)\cos\theta - 4m_1 z\dot{\theta}\dot{z})}{2(m_2 l^2 + 3m_1 z^2)} \end{bmatrix}$$

Then equilibria when $\vec{f}(\vec{x}_e, \vec{u}_e) = \vec{0}$ ...implies...

$\dot{z}_e = 0$

$\dot{\theta}_e = 0$

$-g\sin\theta_e = 0 \Rightarrow \theta_e = \pi n$ for $n \in \mathbb{Z}$

$$\frac{3((2F_e l - m_2 g l - 2m_1 g z)\cos\theta_e)}{2(m_2 l^2 + 3m_1 z_e^2)} = 0 \Rightarrow 2F_e l - m_2 g l - 2m_1 g z_e = 0$$

$\Rightarrow F_e l = \frac{1}{2} m_2 g l + m_1 g z_e$

$\Rightarrow F_e = \frac{1}{2} m_2 g + m_1 g \frac{z_e}{l}$

Therefore equilibria occur at...

$$\left.\begin{array}{ll} \dot{z}_e = 0 & \theta_e = \pi n \text{ for } n \in \mathbb{Z} \\ \dot{\theta}_e = 0 & F_e = \frac{1}{2} m_2 g + m_1 g \frac{z_e}{l} \end{array}\right\} \text{eqba.}$$

All other state variables (z_e) are unconstrained.

Then by solving the jacobian of $\dot{\vec{x}}$ at equilibrium $\vec{x}_e = [z_e, 0, 0, 0]^T$

with $\vec{u}_e = [F_e] = [\frac{1}{2}(m_2 + m_1 g \frac{z_e}{l}]$...

$$\dot{\vec{\tilde{x}}} = \cancel{\begin{bmatrix} \dot{\tilde{z}} \\ \dot{\tilde{\theta}} \\ -g\tilde{\theta} \\ \frac{12\tilde{F}}{l(3m_1 + 4m_2)} - \frac{12 g m_1 \tilde{z}}{l^2(3m_1 + 4m_2)} \end{bmatrix}} = \underbrace{\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -g & 0 & 0 \\ \frac{-3gm_1}{m_2 l^2 + 3m_1 z_e^2} & 0 & 0 & 0 \end{bmatrix}}_{J} \begin{bmatrix} \tilde{z} \\ \tilde{\theta} \\ \dot{\tilde{z}} \\ \dot{\tilde{\theta}} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{3l}{m_2 l^2 + 3m_1 z_e^2} \end{bmatrix} [\tilde{F}]$$

Feedback linearization requires a forcing term to be in every non-linear EoM. However, $-g\sin\theta + z\dot{\theta}^2$ has no forcing term, so feedback linearization is not possible. FL

For state space form, see the <u>Jacobian linearization</u> for the first equation. The second is given (by inspection) as... SSF

$$\vec{\tilde{y}} = \begin{bmatrix} \tilde{z} \\ \tilde{\theta} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} \tilde{z} \\ \tilde{\theta} \\ \dot{\tilde{z}} \\ \dot{\tilde{\theta}} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} [\tilde{F}]$$

<u>Problem F.4, F.6</u>

<u>Find</u>: Same as D, with $\vec{u} = \begin{bmatrix} F \\ \tau \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ d & -d \end{bmatrix} \begin{bmatrix} f_r \\ f_l \end{bmatrix} = \begin{bmatrix} f_r + f_l \\ d(f_r - f_l) \end{bmatrix}$.

This implies $f_r = \frac{1}{2}F + \frac{1}{2d}\tau$ and $f_l = \frac{1}{2}F - \frac{1}{2d}\tau$.

$\vec{x} = [z_v, h, \theta, \dot{z}_v, \dot{h}, \dot{\theta}]^T$ and $\vec{y} = [z_v, h]^T$.

From assignment #3...

$$\dot{\vec{x}} = \vec{f}(\vec{x}, \vec{u}) = \begin{bmatrix} d/dt(z_v) \\ d/dt(h) \\ d/dt(\theta) \\ d/dt(\dot{z}_v) \\ d/dt(\dot{h}) \\ d/dt(\dot{\theta}) \end{bmatrix} = \begin{bmatrix} \dot{z}_v \\ \dot{h} \\ \dot{\theta} \\ \dfrac{-((f_l+f_r)\sin\theta + \mu\dot{z}_v)}{m_c + 2m_r} \\ \dfrac{(f_l+f_r)\cos\theta - g(m_c+m_l+m_r)}{m_c+2m_r} \\ \dfrac{d(f_r-f_l)}{J_c + 2d^2 m_r} \end{bmatrix} = \begin{bmatrix} \dot{z}_v \\ \dot{h} \\ \dot{\theta} \\ \dfrac{-F\sin\theta - \mu\dot{z}_v}{m_c + 2m_r} \\ \dfrac{F\cos\theta - g(m_c+m_l+m_r)}{m_c+2m_r} \\ \dfrac{\tau}{J_c + 2d^2 m_r} \end{bmatrix}$$

Therefore equilibria occur when $\vec{f}(\vec{x}_e, \vec{u}_e) = 0$...

$\dot{z}_{ve} = 0$ $\quad -F_e\sin\theta_e = 0 \Rightarrow F_e = 0$ or $\theta_e = n\pi$ for $n \in \mathbb{Z}$

$\dot{h}_e = 0$ $\quad F_e\cos\theta_e = g(m_c + m_l + m_r) \Rightarrow F_e = \dfrac{g(m_c+m_l+m_r)}{\cos\theta_e} \neq 0$

$\dot{\theta}_e = 0$ $\quad \tau_e = 0$

Therefore equilibria occur for...

$\dot{z}_{ve} = 0$ $\quad \dot{\theta}_e = 0$ $\quad \theta_e = n\pi$ for $n \in \mathbb{Z}$ $\quad z_{ve}, h \in \mathbb{R}$

$\dot{h}_e = 0$ $\quad \tau_e = 0$ $\quad F_e = \cos\theta_e\, g(m_c + m_l + m_r)$ $\quad \forall$ — eqba

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Then selecting point $\vec{x}_e = [0, 0, 0, 0, 0, 0]^T$ and $\vec{u}_e = [\cancel{(m_c+m_l+m_r)g}\ F_e,\ 0]^T$ as the equilibrium to evaluate the Jacobian, sympy yields...

$$\dot{\tilde{\vec{x}}} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & \frac{-F_e}{m_c+2m_r} & \frac{-\mu}{m_c+2m_r} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \tilde{z}_v \\ \tilde{h} \\ \tilde{\theta} \\ \dot{\tilde{z}}_v \\ \dot{\tilde{h}} \\ \dot{\tilde{\theta}} \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \frac{1}{m_c+2m_r} & 0 \\ 0 & \frac{1}{J_c+2d^2m_r} \end{bmatrix} \begin{bmatrix} \tilde{F} \\ \tilde{\tau} \end{bmatrix}$$

JL

Feedback linearization is no possible because there are 2 equations that are non-linear and only one associated input variable, F. Additionally, F is <u>attached</u> to the nonlinear term, which makes it impossible also without setting $F=0$.

FL

For SSF, see <u>JL above for $\dot{\tilde{\vec{x}}}$</u>. By inspection,

$$\vec{y} = \begin{bmatrix} \tilde{z} \\ \tilde{h} \\ \tilde{\theta} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \tilde{z}_v \\ \tilde{h} \\ \tilde{\theta} \\ \dot{\tilde{z}}_v \\ \dot{\tilde{h}} \\ \dot{\tilde{\theta}} \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \tilde{F} \\ \tilde{\tau} \end{bmatrix}$$

Note that rows 2 and 5 give longitudinal solution and rows 1, 3, 4 & 6 give the lateral states. Corresponding columns of A would likewise be replaced.

SSF